

Introduction

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Is β reliable?

Bias: does
 $E(\hat{\beta}|X) = \beta$
hold?

When is the variance
low?

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Example: Light
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ECONOMETRICS I

Lecture 6

What variables to include in a regression?

Bias-efficiency tradeoffs

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Introduction

We want to answer:

- What variables should I include in a regression?
- All possible ones?
- What happens if an important variable is missing?
- What happens if an unimportant one is included?

Thus: perhaps the most important lecture!

Key insights:

- Tradeoff: bias vs. variance.
- Some tools to discriminate.

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In the **univariate** model ($\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x$), the **slope** estimate is

$$\hat{\beta}_2 = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

and its **variance** in repeated sampling is

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$$

Remember our OLS-estimator formulas

In the **multivariate** case ($\hat{y} = \hat{\beta}_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$), the slopes are

$$\underset{k \times 1}{\hat{\beta}} = \begin{bmatrix} \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = (X'X)^{-1}X'y$$

with **variance-covariance matrix**

$$\underset{k \times k}{V(\hat{\beta}|X)} = \begin{bmatrix} \text{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \text{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2(X'X)^{-1}.$$

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IS $\hat{\beta}$ RELIABLE?

Is $\hat{\beta}$ a reliable estimator?

■ Unbiased?

- Will estimate $\hat{\beta}$ be on average like true β ?
- Ideally: $\mathbb{E}(\hat{\beta}) = \beta$

■ Efficient?

- Will estimate $\hat{\beta}$ be usually close to β ?
- Ideally: **small $\text{var}(\hat{\beta})$**

Bias: does $E(\hat{\beta}|X) = \beta$ hold?

Is our set of estimates $\hat{\beta}_1 \dots \hat{\beta}_k$ unbiased?

$$\hat{\beta} = (X'X)^{-1}X'y$$

$$= (X'X)^{-1}X'(X\beta + u) \quad \text{given } \boxed{y = X\beta + u}$$

$$= \beta + (X'X)^{-1}X'u$$

$$\mathbb{E}(\hat{\beta}|X) = \beta + \mathbb{E}[(X'X)^{-1}X'u|X]$$

$$= \beta + (X'X)^{-1}X'\mathbb{E}[u|X]$$

$$= \beta \quad \text{given } \boxed{\mathbb{E}[u|X] = 0}$$

Unbiasedness may not hold if:

- true data generating process differs from the estimated model $y = X\beta + u$
- $\text{cov}(x_j, u) \neq 0$, for some $j \in \{1, \dots, k\}$, violating assumption $\mathbb{E}[u|X] = 0$

Bias: does $E(\hat{\beta}|X) = \beta$ hold?

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Thus, unbiasedness relies on key assumptions:

- 1 **Correct specification.** Unobserved reality behaves like

$$y = X\beta + u.$$

- 2 **Exogenous errors.** The error has zero mean given *any* value of X : $\mathbb{E}(u|X) = 0$. This implies $\text{cov}(x_j, u) = 0$ for all $j = 1 \dots k$.

When is the variance low?

We obtained the variance-covariance matrix

$$V(\hat{\beta}|X) = \begin{bmatrix} \text{var}(\hat{\beta}_1|X) & & \\ & \ddots & \\ & & \text{var}(\hat{\beta}_k|X) \end{bmatrix} = \sigma^2(X'X)^{-1}$$

$k \times k$

For $k = 2$ (univariate regression), we know that the slope variance is

$$\text{var}(\hat{\beta}_2|x) = \frac{\sigma^2}{\sum (x_i - \bar{x})^2} = \frac{\sigma^2}{n \text{ var}(x)}$$

which declines with $\sigma^2 \downarrow$ and $n \uparrow$.

Key questions:

- What's the variance of any coefficient $\hat{\beta}_j$ when $k > 2$?
- Does $\text{var}(\hat{\beta}_j|X)$ change as we include/exclude other variables?

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FRISCH-WAUGH-LOVELL (FWL) THEOREM

Frisch-Waugh-Lovell (FWL) Theorem

The **Frisch-Waugh-Lovell (FWL) Theorem** provides an extremely useful tool to study $\mathbb{E}(\hat{\beta}_j)$ and $\text{var}(\hat{\beta}_j)$ as a function of other included variables.

Key ingredients

- **Auxiliary regression** of x_j w.r.t. all other regressors $x_{\neq j}$

$$x_j = X_{\neq j}\hat{\gamma} + \underset{\text{residuals}}{\tilde{x}_j}$$

- **Residuals** $\tilde{x}_j$ of that auxiliary regression
- **Fit-goodness** R_j^2 of that of that auxiliary regression

Frisch-Waugh-Lovell (FWL) Theorem

Theorem claim 1:

Estimating the OLS coefficient $\hat{\beta}_j$ through the multiple regression

$$y = X\hat{\beta} + \hat{u},$$

or the the OLS coefficient $\tilde{\beta}_j$ through the simple regression

$$\hat{y} = \tilde{\beta}_1 + \tilde{\beta}_j \tilde{x}_j + \tilde{u},$$

(with $\tilde{x}_j$ = residuals from auxiliary regression) yields the same

$$\boxed{\tilde{\beta}_j = \hat{\beta}_j} \text{ and } \boxed{\text{var}(\tilde{\beta}_j) = \text{var}(\hat{\beta}_j)}.$$

Frisch-Waugh-Lovell (FWL) Theorem

Example:

	Dependent variable :		
	income2010 (1)	ineq1950 (2)	income2010 (3)
ineq1950	-16,181.260*** (4,466.854)		
income1950	0.844 (0.552)	-0.0001 (0.00003)	
xtilde			-16,181.260** (6,546.118)
Constant	49,298.500*** (10,741.530)	2.330*** (0.179)	20,239.130*** (1,809.628)
Observations	13	13	13

Note equal estimate $-16,181.260$. Warning: The reported standard errors in (3) are misleading; in practice real variance of the coefficient in (1) and (3) must obviously coincide.

Frisch-Waugh-Lovell (FWL) Theorem

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**Frisch-Waugh-Lovell
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Theorem claim 2:

The variance of any regressor $j \in \{1, \dots, k\}$ is

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2}$$

where R_j^2 is the R^2 of the auxiliary regression of x_j against all other regressors.

Where does claim 2 come from?

Consider: $y = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + u$

Auxiliary regression: $x_2 = \delta_1 + \delta_2 x_3 + \tilde{x}_2$ yields
Dep. var **residual**

$$R_{\text{aux}}^2 = 1 - \frac{\text{var}(\tilde{x}_2)}{\text{var}(x_2)}, \text{ hence } \text{var}(\tilde{x}_2) = (1 - R_{\text{aux}}^2) \text{var}(x_2)$$

Following claim 1 of the FWL theorem

$$\hat{\beta}_2 = \tilde{\beta}_2 = \frac{\text{cov}(\tilde{x}_2, y)}{\text{var}(\tilde{x}_2)}$$

Thus:

$$\text{var}(\hat{\beta}_2|X) = \frac{\sigma^2}{n \text{var}(\tilde{x})} = \frac{\sigma^2}{n(1 - R_{\text{aux}}^2) \text{var}(x)}$$

More generally

For x_j with k total regressors:

- Regress x_j on all other x 's:

$$x_j = \delta_0 + \delta_1 x_1 + \cdots + \delta_{j-1} x_{j-1} + \delta_{j+1} x_{j+1} + \cdots + \delta_k x_k + \tilde{x}_j$$

residual

- Get R_j^2 from this regression

- Residual variance: $\sum \tilde{x}_{ji}^2 = (1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2$

Thus:

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{(1 - R_j^2) \sum (x_{ji} - \bar{x}_j)^2}$$

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Interpretation

Rewriting in compact notation:

$$\text{var}(\hat{\beta}_j) = \frac{\sigma^2}{n} \cdot \frac{1}{1 - R_j^2} \cdot \frac{1}{\text{var}(x_j)}$$

Where:

- $\frac{\sigma^2}{n}$: Basic sampling variance
- $\frac{1}{\text{var}(x_j)}$: More variable $x \Rightarrow$ more precision
- $\frac{1}{1 - R_j^2}$: **Variance Inflation Factor (VIF)**

Interpretation: When x_j is highly predictable from other x 's ($R_j^2 \rightarrow 1$), $\text{VIF} \rightarrow \infty$ and variance explodes!

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OMITTED RELEVANT VARIABLES

Population Model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \quad (E[\varepsilon|x_1, x_2] = 0)$$

Estimated Model:

$$y = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{u}$$

Problem: Now our error term is $u = \beta_2 x_2 + \varepsilon!!!$
 → Exogeneity assumption $\text{cov}(x_1, u) = 0$ invalidated when:

- $\beta_2 \neq 0$ (causal effect exists)
- $\text{cov}(x_1, x_2) \neq 0$ (variables correlated)

Omitted variable bias (OVB)

Consider the case $x_2 = \delta x_1 + \nu$.

We estimate

$$\begin{aligned}\hat{\beta}_1 &= \frac{\text{cov}(x_1, y)}{\text{var}(x_1)} \\ &= \frac{\text{cov}(x_1, \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon)}{\text{var}(x_1)} \\ &= \beta_1 + \frac{\text{cov}(x_1, \beta_2 x_2 + \varepsilon)}{\text{var}(x_1)} \\ &= \beta_1 + \beta_2 \frac{\text{cov}(x_1, x_2)}{\text{var}(x_1)} + \frac{\text{cov}(x_1, \varepsilon)}{\text{var}(x_1)}\end{aligned}$$

$$E[\hat{\beta}_1|X] = \beta_1 + \beta_2 \delta$$

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Omitted Variable Bias

- True model: $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$
- Estimated model: $y = \hat{\beta}_1 x_1 + \hat{u}$
- Bias: $E[\hat{\beta}_1] = \beta_1 + \beta_2 \delta$
where δ from $x_2 = \delta x_1 + v$
- **Bias occurs when:**
 1. $\beta_2 \neq 0$ (omitted variable matters)
 2. $\delta \neq 0$ (x_2 correlates with x_1)

Example: Light Exposure and Myopia Development

Conflicting Studies:

- **UPenn (1999, Nature):** Artificial light exposure $\uparrow$ myopia risk in infants
- **Ohio State (2000, Nature):** No significant effect of light exposure

Variable Mapping:

- y : Future myopia development
- X_1 : Light exposure + controls
- X_2 : Parental myopia (omitted in UPenn study)
- $\beta_2 > 0$: Genetic transmission exists
- $\delta > 0$: Myopic parents tend to leave lights on

Example: Light Exposure and Myopia Development

OV-Bias

$$E[\hat{\beta}_{light}] = \beta_{light} + \underbrace{\delta \cdot \beta_2}_{Bias > 0}$$

- $\beta_2 > 0$ (genetic transmission exists)
- $\delta > 0$ (myopic parents use more light)
- $\Rightarrow$ **Positive bias** in light exposure coefficient

Consequences:

- **Overestimated** t-statistic: $t_{light} = \frac{\hat{\beta}_{light}}{\sqrt{\widehat{var}}}$
- **Underestimated** p-value
- **Exaggerated** significance
- UPenn: Found spurious significance due to omitted variable bias

Parental myopia was the confounding variable explaining both light exposure patterns and child myopia.

Variance Analysis

Variance (by FWL):

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{n} \frac{1}{(1 - R_{x_1, x_2}^2)} \frac{1}{\text{var}(x_1)}$$

where:

- R_{x_1, x_2}^2 from regressing x_1 on x_2
- σ^2 estimated with $\hat{\sigma}^2 = \frac{SSR}{n-k}$

Key Insight: Omitting x_2 :

- **Reduces** denominator $(1 - R^2) \downarrow$ (lowers variance)
- But introduces **bias** $\beta_2\delta$

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INCLUDED IRRELEVANT VARIABLES

Population Model:

$$y = \beta_1 x_1 + u \quad (E[u|x_1] = 0)$$

Estimated Model:

$$y = \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{u}$$

Now $\text{cov}(u, x_1) = 0$ holds!

Is this better? Should we try to include as many variables as possible?

Bias & Variance

Unbiasedness Preservation (since now $\text{cov}(u, x) = 0$ holds)

$$E[\hat{\beta}_1] = \beta_1 \quad (\text{when } E[u|x_1, x_2] = 0)$$

Variance Inflation:

$$\text{var}(\hat{\beta}_1) = \frac{\sigma^2}{n} \frac{1}{(1 - R_{x_1, x_2}^2)} \frac{1}{\text{var}(x_1)}$$

- $R_{x_1, x_2}^2 > 0$ inflates variance via VIF
- And since σ^2 must be estimated via $\hat{\sigma}^2 = \sum \hat{u}_i^2 / (n - k)$:
No. of regressors $k \uparrow \rightarrow \hat{\sigma}^2 \uparrow$, especially when n is low.

Bias & Variance

Redundancy & Variance Inflation

- VIF formula: $VIF = \frac{1}{1 - R^2_{x_j|x_{-j}}}$

where $R^2_{x_j|x_{-j}}$ comes from regressing x_j on other predictors

- Interpretation:
 - $R^2 = 0.75 \Rightarrow VIF = 4$ ($4 \times$ variance inflation)
 - $R^2 = 0.99 \Rightarrow VIF = 100$ (severe multicollinearity)

Problem: Including irrelevant x_2 when:

- $\beta_2 = 0$ (no causal effect)
- $\text{cov}(x_1, x_2) \neq 0$ (variables correlated)

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What VIF Measures

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Auxiliary regression: Regress x_1 on all other predictors
($x_2, x_3, \dots$)

- R^2_{aux} close to 0: Other predictors don't explain x_1 well
 - $\text{VIF} \approx 1$ (no inflation)
 - Variance stays small
- R^2_{aux} close to 1: Other predictors explain x_1 very well
 - VIF becomes large
 - Variance gets inflated

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PRACTICAL IMPLICATIONS

Bias-Variance Tradeoff

■ Omitted variables:

Bias: $\beta_2\delta$

Variance: May $\downarrow$ (if x_2 noisy) or $\uparrow$ (if x_2 informative)

■ Redundant variables:

Bias: None

Variance: $\uparrow$ via VIF

■ Practical dilemma:

- Overspecification $\Rightarrow$ efficiency loss
- Underspecification $\Rightarrow$ potential bias
- Solution: Think about really relevant variables. **Avoid redundancy** but **include possible confounders** of the treatment.

Bias-Variance Tradeoff:

	Omitted Relevant	Included Irrelevant
Bias	$\beta_2\delta \neq 0$	0
Variance	Potentially $\downarrow$	$\uparrow$ via VIF
Efficiency	May improve	Always worse

Practical examples

Consider the following data generating process

```
set.seed(123) # optional, for exact reproducibility

# Parameters to play with
n <- 50          # Number of observations
sigma2 <- 1      # Error variance
beta1 <- 2       # True coefficient for x1
beta2 <- -2      # True coefficient for x2 (set to 0 for irrelevant variable)
delta <- 0.9     # Relationship between x1 and x2 (cov(x1,x2)/var(x1))

# Generate data
x1 <- rnorm(n)
x2 <- delta*x1 + rnorm(n, sd = sqrt(1 - delta^2)) # Ensure var(x2)=1
u <- rnorm(n, sd = sqrt(sigma2))
y <- beta1*x1 + beta2*x2 + u
irrelevant <- 1.2*x1 + .1*rnorm(n) # irrelevant but correlated with x1
```

Let the “unknown” parameter of interest be $\beta_1 = 2$.
Notice that the irrelevant variable is correlated with x_1

Practical examples

	Omitted (1)	Correct (2)	Overfit (3)
x1	0.262 (0.212)	2.375*** (0.355)	2.769 (1.961)
x2		-2.389*** (0.361)	-2.401*** (0.370)
Irrelevant			-0.323 (1.579)
Observations	50	50	50
R2	0.031	0.498	0.498
Note : *p<0.1; **p<0.05; ***p<0.01			

Results:

- In (1), omitting x_2 biases $\hat{\beta}_1 \downarrow$ (via $\delta\beta_2$). (2) performs better.
- In (3): $VIF \uparrow \rightarrow \text{var}(\hat{\beta}_1) \uparrow \rightarrow \hat{\beta}_1$ not statistically significant.

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Now change to

```
beta2 <- -0.02 # True coefficient for x2 (set to 0 for irrelevant variable)
```

where β_2 continues to be relevant bus has a lower impact on the outcome y ($\beta_2 = -0.02$).

Practical examples

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	Omitted (1)	Correct (2)	Overfit (3)
x1	2.013*** (0.154)	2.375*** (0.355)	2.769 (1.961)
x2		-0.409 (0.361)	-0.421 (0.370)
Irrelevant			-0.323 (1.579)
Observations	50	50	50
R2	0.780	0.786	0.786

Note :

*p<0.1; **p<0.05; ***p<0.01

Despite being incorrect, the **model with the omitted variable performs better!**

Why? You should be able to answer ;-)

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DIAGNOSIS

The Adjusted R^2

Recall the definition of the R^2

$$R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Because R^2 is a function of $\sum_{i=1}^n \hat{u}_i^2$, we have:

$$R^2 \xrightarrow{k \rightarrow \infty} 1$$

We cannot use R^2 to compare the goodness of fit of models with different numbers of regressors, due to a systematic bias where models with high k tend to have higher R^2 .

The Adjusted R^2

A correction for the bias introduced by k is to normalize the coefficient by the degrees of freedom of $\sum_{i=1}^n \hat{u}_i^2$ and $\sum_{i=1}^n (y_i - \bar{y})^2$, which are $n - k$ and $n - 1$ respectively. The resulting statistic is called the **adjusted coefficient of determination** or corrected R^2 :

$$\begin{aligned}\bar{R}^2 &= 1 - \frac{\sum_{i=1}^n \hat{u}_i^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \frac{(n - 1)}{(n - k)} \\ &= 1 - (1 - R^2) \frac{n - 1}{n - k}\end{aligned}$$

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‘Unknown’ data generating process:

```
# Define coefficients
```

```
b1 <- 2
```

```
b2 <- 1.5
```

```
b3 <- -0.05
```

```
b4 <- 0.0005
```

```
b5 <- 0.0002
```

```
# Generate random uniform x values (between 10 and 100)
```

```
z <- runif(n, 10, 100)
```

```
# Data generating process (unknown to the researcher)
```

```
y4 <- b1 + b2*z + b3*z^2 + b4*z^3 + b5*z^4 + rnorm(n, 0, 200)
```


Application to polynomial regression

Then 'naively' run polynomial regression up to order 7

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
z1	187.3132*** (10.0725)	-258.8695*** (11.7546)	92.8726*** (13.4164)	-15.9799 (38.0644)	-40.9772 (88.0659)	49.4034 (210.4774)	474.7408 (483.2892)
z2		4.0506*** (0.1043)	-3.2942*** (0.2699)	0.3900 (1.2397)	1.6062 (4.0556)	-4.1634 (12.8549)	-37.5053 (36.4448)
z3			0.0444*** (0.0016)	-0.0043 (0.0161)	-0.0306 (0.0851)	0.1449 (0.3808)	1.4582 (1.3961)
z4				0.0002*** (0.0001)	0.0005 (0.0008)	-0.0023 (0.0059)	-0.0308 (0.0297)
z5					-0.000001 (0.000003)	0.00002 (0.00004)	0.0004 (0.0004)
z6						-0.000000 (0.000000)	-0.000002 (0.000002)
z7							0.0000 (0.0000)
Constant	-5,965.6510*** (614.6885)	3,721.8920*** (292.0271)	-758.4058*** (190.7420)	243.0720 (376.9863)	420.7805 (679.3492)	-98.3478 (1,291.8580)	-2,176.1190 (2,487.1100)
Observations	100	100	100	100	100	100	100
R2	0.7792	0.9867	0.9985	0.9986	0.9986	0.9986	0.9986
Adjusted R2	0.7769	0.9864	0.9984	0.9986	0.9986	0.9985	0.9985

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Notice that $\bar{R}^2$ begins to fall after order 4.

Alternative intra-sample indicators that you may learn in
Econometrics II:

- AIC (Akaike Information criterion)
- BIC (Bayes Information criterion)
- Hannan-Quinn criterion

Ideally: Out-of-Sample Evaluation

The ideal criterion to judge model quality is to examine its performance “out-of-sample” using data **not used** for estimation.

Procedure:

- 1 Estimate models with n observations
- 2 Evaluate performance with n_p prediction data points

Requirement:

- 1 Large total no. of observations: $N = n + n_p$

Ideally: Out-of-Sample Evaluation

We examine prediction errors $e_i = y_i - \hat{y}_i$. Common evaluation metrics:

1 Mean Squared Error (MSE) / RMSE:

$$\text{RMSE} = \left(\frac{1}{n_p} \sum (y_i - \hat{y}_i)^2 \right)^{\frac{1}{2}}$$

2 Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n_p} \sum |y_i - \hat{y}_i|$$

Where n_p = number of predicted observations.

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TAKEAWAYS

Key takeaways

You should now know:

- What is the omitted variable bias (OVB)
- What the OVB means for the assumption $\mathbb{E}[u_i|X] = 0$
- The impact of the OVB on $\mathbb{E}(\hat{\beta}_j|X)$ and $\text{var}(\hat{\beta}_j|X)$
- The consequences of including irrelevant variables
- The nature of the bias-efficiency tradeoff
- The FWL theorem
- Diagnosis through intra-sample (Adjusted- R^2) or out-of-sample (e.g. MSEP) indicators.

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